UT 2 – GUI with Jetpack Compose

Warm-up exercises · Blocks 1 and 2

Interface Development (0488) · 2nd DAM · 2026/2027

How to work

These ten exercises help you to put together what you have learnt in Block 1 (composables, `Text`, modifiers, previews) and Block 2 (`Column`, `Row`, `Box`, `Arrangement`, `Alignment`, size, `weight`, colour and shape). They go from very simple to a complete small screen, one small step at a time, so that you get used to Android Studio and to the Preview.

> **No emulator needed**
> Every exercise can be solved and checked **only with the Preview**. Do not run the app on the emulator: compare the Preview with the image of the exercise.

Rules

- Work in your `UT2_Compose` project, in the package `com.example.myfirstcomposeapp.exercises.warmup`.
- Use one file per exercise: `Warmup01.kt`, `Warmup02.kt`... The main composable of each exercise is called `Warmup01`, `Warmup02`... and its preview `Warmup01Preview`, `Warmup02Preview`...
- Every preview must use `@Preview(showBackground = true)` and wrap the content in `MyFirstComposeAppTheme`, unless the exercise says otherwise.
- The images are a guide: sizes are in `dp` and `sp`, and colours are given in hexadecimal (for example `Color(0xFF1565C0)`). Your result should look the same, not necessarily pixel by pixel.
- Work in small steps: write a little, look at the Preview, and fix. Commit when an exercise works.
- Try each exercise on your own first. If you get stuck, read the hints and the sections of the documents that are indicated, and then ask.

Overview

No.	Exercise	You practise	Sections
1	Welcome text	`Text` style, `padding`	1.7
2	Price tag	Composable with parameters, `clip`, `background`, several previews	1.7, 1.8, 2.8
3	Contact card	`Column`, `spacedBy`, `Alignment`	2.2, 2.5
4	Navigation bar	`Row`, `Arrangement`, `Alignment`	2.3, 2.5
5	Notification badge	`Box`, `align`, `CircleShape`	2.4, 2.8
6	Day bar	`weight`, `clip`, `Column` and `Row` together	2.6, 2.8

No.	Exercise	You practise	Sections
7	Profile card	Order of modifiers, `border`, `Row` + `Column` + `Box`	2.7, 2.8
8	Screen skeleton	Areas with `weight`, `Spacer`, `fillMaxSize`	2.6, 2.7
9	Stat cards	Reusable composable with `modifier`	2.10
10	Shopping list	Data, `forEachIndexed`, composables	2.7, 2.10, 2.11

Exercise 1 · Welcome text

Level 1 of 5 · Warm-up

Show a welcome text with a style, inside a composable.



Target for Exercise 1

What you must build

- A composable `Warmup01` with a single `Text` that shows “Welcome to Compose”.
- Font size `28.sp`, bold, colour `Color(0xFF1565C0)`.
- A space of `16.dp` around the text.
- A preview with a white background.

Hints

- Use the parameters `fontSize`, `fontWeight` and `color` of `Text`, and `Modifier.padding(...)`.

Exercise 2 · Price tag

Level 1 of 5 · Warm-up

Create a small reusable composable and see it with different data.



Target for Exercise 2: three previews of the same composable

What you must build

- A composable `PriceTag(price: String, modifier: Modifier = Modifier)`.
- It shows the price in bold white text of size `18.sp`, on an orange background `Color(0xFFEF6C00)` with rounded corners of `8.dp`.
- Space inside the tag: `12.dp` on the sides and `6.dp` above and below.
- Three previews with the texts “9.99 €”, “120 €” and “0.50 €”.

Hints

- The modifiers go in this order: the `modifier` parameter first, then `clip`, `background` and `padding` (Block 1, section 1.7 and Block 2, section 2.8).
- If the corners are not rounded, check the order of `clip` and `background`.

Exercise 3 · Contact card

Level 2 of 5 · Layout basics

Place three texts one below the other and centre them.



Target for Exercise 3

What you must build

- A composable `Warmup03` with a `Column` that takes all the width and has a light grey background `Color(0xFFF5F5F5)` and `16.dp` of space inside.
- Three texts: “Ana García” (`22.sp`, bold), “Android developer” (`16.sp`, grey `Color(0xFF616161)`) and “Villarrobledo” (`14.sp`, same grey).
- A gap of `4.dp` between the texts, and all of them centred horizontally.

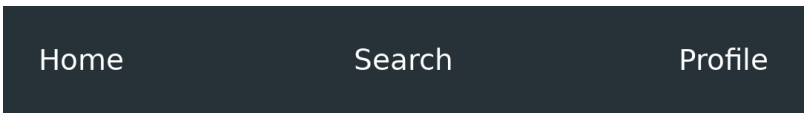
Hints

- `verticalArrangement = Arrangement.spacedBy(...)` for the gaps and `horizontalAlignment` for the centring (section 2.5).
- To centre the texts on the screen, the column must take the whole width. Which modifier does that?

Exercise 4 · Navigation bar

Level 2 of 5 · Layout basics

Place three labels in a row and distribute them.



Target for Exercise 4

What you must build

- A composable `Warmup04` with a `Row` that takes all the width, with a dark background `Color(0xFF263238)`.
- Space inside the bar: `16.dp` on the sides and `12.dp` above and below.
- Three white texts of size `16.sp`: “Home”, “Search” and “Profile”. The first one at the left end, the last one at the right end and the other one in the middle.
- The texts are centred vertically.

Hints

- Look at the table of values of `Arrangement` in section 2.5.
- Then change the arrangement to `SpaceEvenly` and write in a comment what changes.

Exercise 5 · Notification badge

Level 3 of 5 · Layering and sharing

Place one element over another and give a circular shape to a small box.



Target for Exercise 5

What you must build

- A composable `Warmup05` with a `Box` of `120.dp` × `120.dp`, light grey `Color.LightGray`, with rounded corners of `16.dp`.
- In the centre, the text “Mail” (`20.sp`, bold).
- In the top-right corner, a red circle of `28.dp` (`Color(0xFFE53935)`) with the text “3” (white, bold, `14.sp`) centred inside it. Leave `6.dp` between the circle and the edges of the grey square.

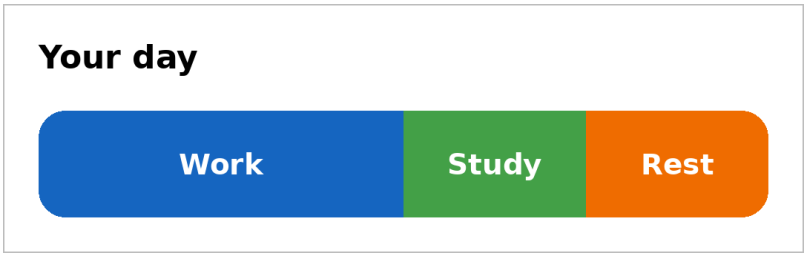
Hints

- The grey `Box` needs `contentAlignment` for the centred text; the circle needs `Modifier.align(...)` (section 2.4).
- `CircleShape` is used with `clip`.
- A `Box` inside another `Box`: each one has its own `contentAlignment`.
- The space of `6.dp` is a `padding` that goes **before** the `size` of the circle.

Exercise 6 · Day bar

Level 3 of 5 · Layering and sharing

Share the width of a row among three boxes and combine a `Column` with a `Row`.



Target for Exercise 6

What you must build

- A composable `Warmup06` with a `Column` that takes all the width, with `16.dp` of space inside and a gap of `8.dp` between its children.
- First child: the text “Your day” (`18.sp`, bold).
- Second child: a `Row` of `48.dp` of height with three boxes that share the width: “Work” (blue `Color(0xFF1565C0)`) takes half, “Study” (green `Color(0xFF43A047)`) and “Rest” (orange `Color(0xFFEF6C00)`) take a quarter each.
- The texts of the boxes are white, bold, `16.sp` and centred. The whole bar has rounded corners of `12.dp`.

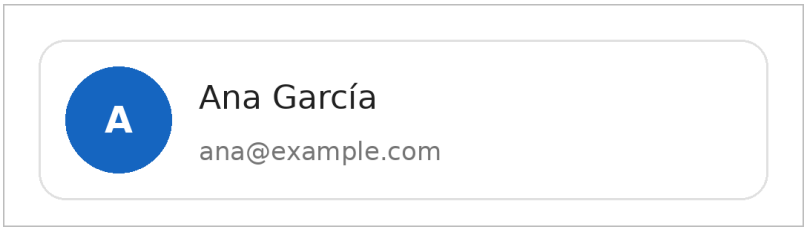
Hints

- Use `weight(2f)`, `weight(1f)` and `weight(1f)` (section 2.6). The boxes also need `fillMaxHeight()`.
- The rounded corners are applied with `clip` to the **`Row`**, not to every box.

Exercise 7 · Profile card

Level 4 of 5 · Combining layouts

Build a card with an avatar and two lines of text, and care about the order of the modifiers.



Target for Exercise 7

What you must build

- A composable `Warmup07(modifier: Modifier = Modifier)` with a `Row` that takes all the width, white background, rounded corners of `12.dp` and a border of `1.dp` of colour `Color(0xFFE0E0E0)` with the same shape.
- `12.dp` of space inside the card, the content centred vertically and `12.dp` between the avatar and the texts.
- Avatar: a blue circle of `48.dp` (`Color(0xFF1565C0)`) with the letter "A" (white, bold, `20.sp`) in the centre.
- Next to the avatar, a `Column` with "Ana García" (`18.sp`) and "ana@example.com" (`14.sp`, grey `Color(0xFF757575)`).
- In the preview, call the composable with `Modifier.padding(16.dp)`, so that the card does not touch the edges.

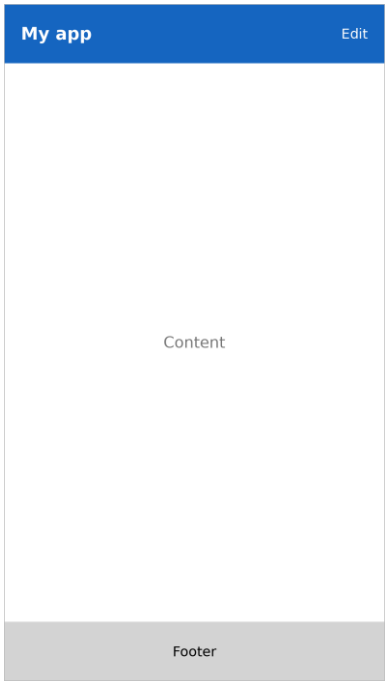
Hints

- Order for the card: size, `clip`, `background`, `border`, `padding` (section 2.8).
- The tree is: `Row` → (`Box` → `Text`) and (`Column` → `Text`, `Text`). Draw it before you write the code.

Exercise 8 · Screen skeleton

Level 4 of 5 · Combining layouts

Divide a whole screen into three areas and push one element to the end of a row.



Target for Exercise 8

What you must build

- A composable `Warmup08` with a `Column` that fills the whole screen, with three areas.
- **Header:** a `Row` that takes the width, blue background `Color(0xFF1565C0)`, `16.dp` inside, content centred vertically. “My app” at the left (white, bold, `20.sp`) and “Edit” at the right end (white, `16.sp`).
- **Content:** an area that takes all the remaining height and has the text “Content” (`18.sp`, grey `Color(0xFF757575)`) in the centre.
- **Footer:** a box of `56.dp` of height, light grey, with the text “Footer” in the centre.
- Preview with `@Preview(showBackground = true, widthDp = 360, heightDp = 640)`.

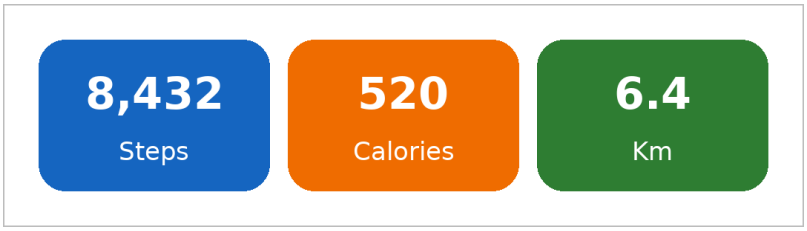
Hints

- You do not need to calculate any height for the content: use `weight(1f)` in the `Column` (section 2.6).
- To push “Edit” to the end of the header, use a `Spacer` with `weight(1f)` between the two texts.

Exercise 9 · Stat cards

Level 5 of 5 · Reuse and data

Create a reusable composable and let its caller decide how it fits in a row.



Target for Exercise 9

What you must build

- A composable `StatCard(title: String, value: String, color: Color, modifier: Modifier = Modifier)`: a card with rounded corners of `12.dp`, background `color`, `12.dp` of space inside, with the `value` (white, bold, `24.sp`) and below it the `title` (white, `14.sp`), both centred.
- A composable `Warmup09` with a `Row` that takes all the width, with `16.dp` of space inside and `8.dp` between cards.
- Three cards of the **same width**: “Steps” “8,432” (blue `Color(0xFF1565C0)`), “Calories” “520” (orange `Color(0xFFEF6C00)`) and “Km” “6.4” (green `Color(0xFF2E7D32)`).
- Two previews: one of a single `StatCard` and one of `Warmup09`.

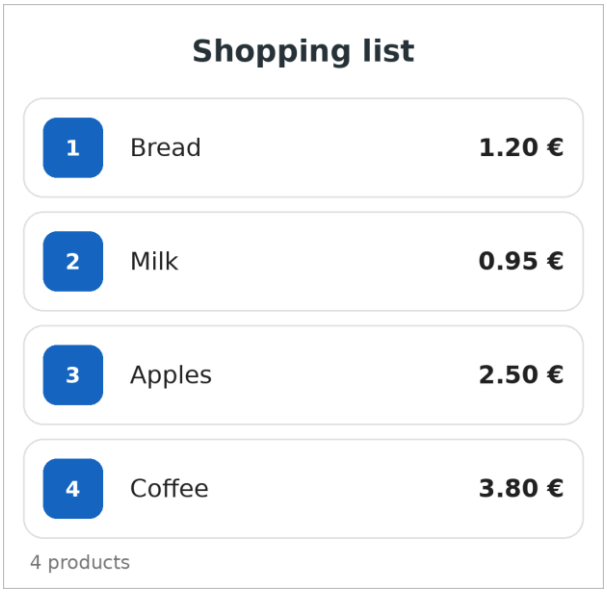
Hints

- `StatCard` cannot use `weight` by itself: the `Row` passes `Modifier.weight(1f)` as the `modifier` parameter (section 2.10).
- Inside `StatCard`, the `modifier` that you receive goes **first** in the chain.

Exercise 10 · Shopping list

Level 5 of 5 · Reuse and data

Integrate everything: a screen built from a list of data, with one composable per row.



Target for Exercise 10

What you must build

- A `data class Product(val name: String, val price: String)` and a list with four products: Bread “1.20 €”, Milk “0.95 €”, Apples “2.50 €” and Coffee “3.80 €”.
- A composable `ProductRow(number: Int, product: Product, modifier: Modifier = Modifier)`: a card like the one of Exercise 7 (white, rounded corners of `12.dp`, border, `12.dp` inside, content centred vertically, `12.dp` between elements) with, from left to right: a blue rounded square of `36.dp` (corners of `8.dp`) with the number in white and bold; the name (`18.sp`); and the price (`18.sp`, bold) at the end of the row.
- A composable `Warmup10` with a `Column` that fills the screen, with background `Color(0xFFFAFAFA)`, `12.dp` of space inside and `8.dp` between its children, with: a title “Shopping list” (`22.sp`, bold, dark `Color(0xFF263238)`) centred; one `ProductRow` for each product, **generated from the list**; and at the end a text with the number of products, for example “4 products” (`14.sp`, grey), written with a string template and `products.size`.
- Preview with `@Preview(showBackground = true, widthDp = 360, heightDp = 640)`.

Hints

- Do not write four `ProductRow` by hand: use `products.forEachIndexed { index, product -> ... }` (section 2.11). The number to show is `index + 1`.
- For the price at the end of the row, use a `Spacer` with `weight(1f)` between the name and the price.
- Change the list (add or remove a product) and check that the Preview updates and the text at the end changes.

Final checklist

Before you finish an exercise, check these points:

- The Preview looks like the image and shows no error.
- The names of the files, composables and previews follow the rules of this document.
- Every composable that is placed by a parent has a `modifier` parameter with a default value, and it is the first modifier in the chain.
- You use `spacedBy` or `Spacer` for gaps between elements, and `weight` to share space, instead of fixed sizes that do not adapt.
- You can explain every line of your code.